

JKBOSE 2026

CLASS 11TH

Morphology of Flowering Plants

LECTURE-03

Biology

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Topics To Covered

1

MORPHOLOGY OF STEM

2

MODIFICATIONS OF STEM

→ Not in JKBose
→ included in NEET Syllabus.

3

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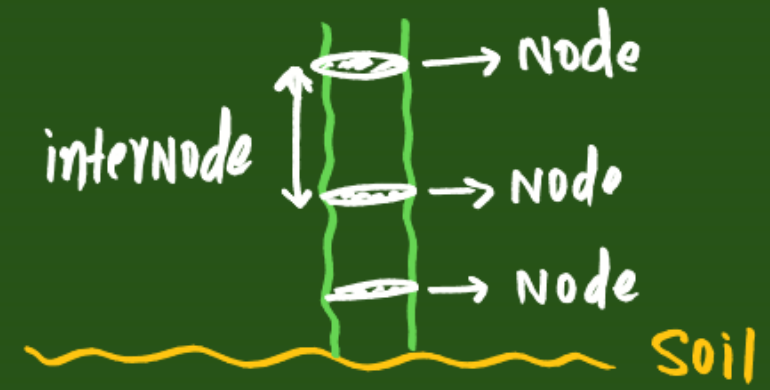
MORPHOLOGY OF STEM

- Stem lies above the Soil.
- It develops from plumule of Embryo
- Stem bears **Nodes** and **internodes**

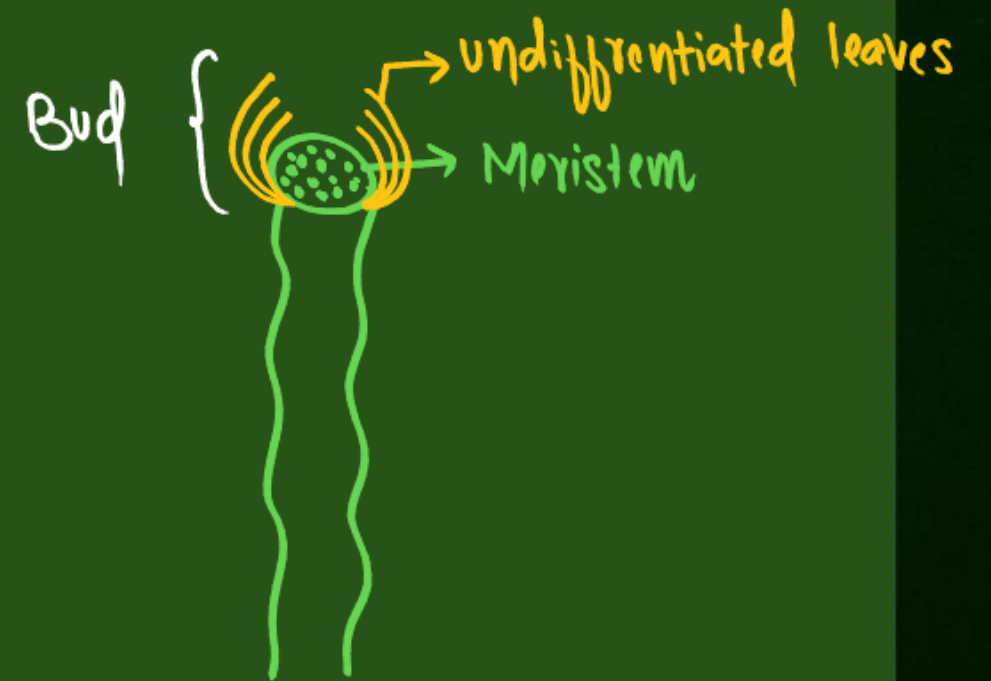
At node - leaves, flowers, branches, buds are present.

- **Stem + Branches** — includes Shoot of plant.

- Stem bears **buds** → Meristematic tissue + undifferentiated leaves



Distance between two consecutive nodes.



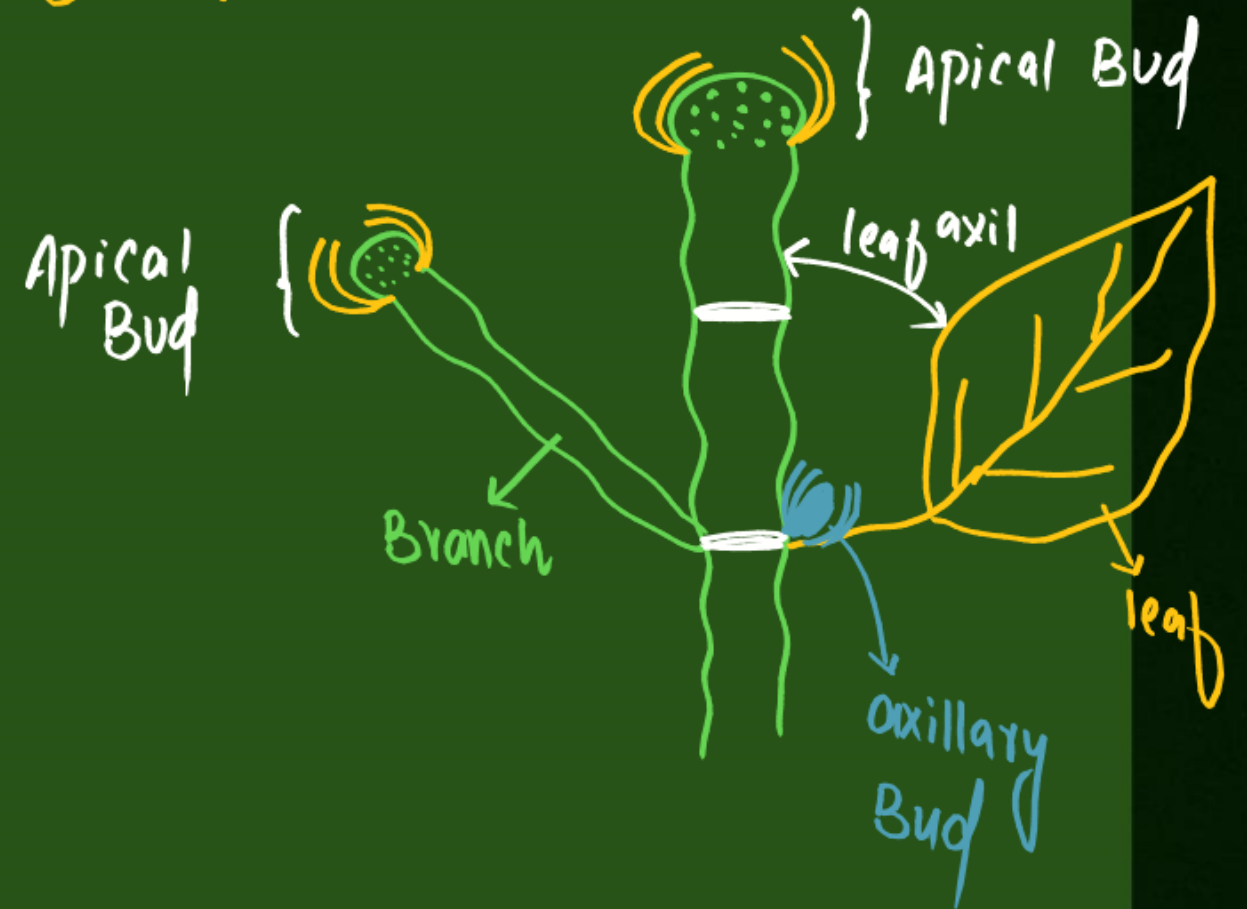
→ Bud is of two types

- ① **Apical Bud** or Terminal Bud → Present at apex / tip of main Stem or Branch.

② **Axillary Bud**

Present in leaf axil.

→ Stem is green when young and later becomes Woody and dark brown.



MAIN FUNCTIONS OF STEM

- ① It bears branches, leaves and flowers.
- ② It helps in transport of H_2O , minerals and food (photosynthates).
- ③ It provides mechanical support to plant.
- ④ Some stems help in vegetative propagation.

MODIFICATIONS OF STEM

- Not in JKBOSE Syllabus.
- Not in NEET Syllabus.

• Some Stems gets modified to perform functions other than main role.

• Stem — 03 types

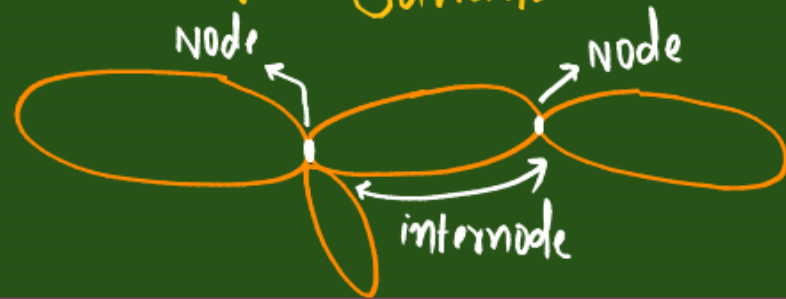
- 1. Underground Stem
- 2. Sub-aerial Stem.
- 3. Aerial / Above ground Stem.



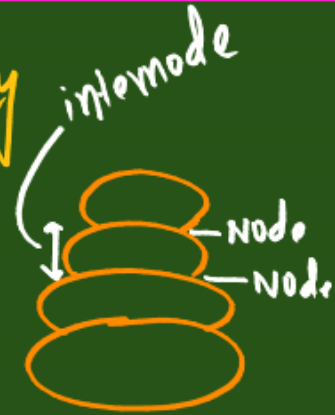
(a) UNDERGROUND STEM MODIFICATIONS

• 04 types :

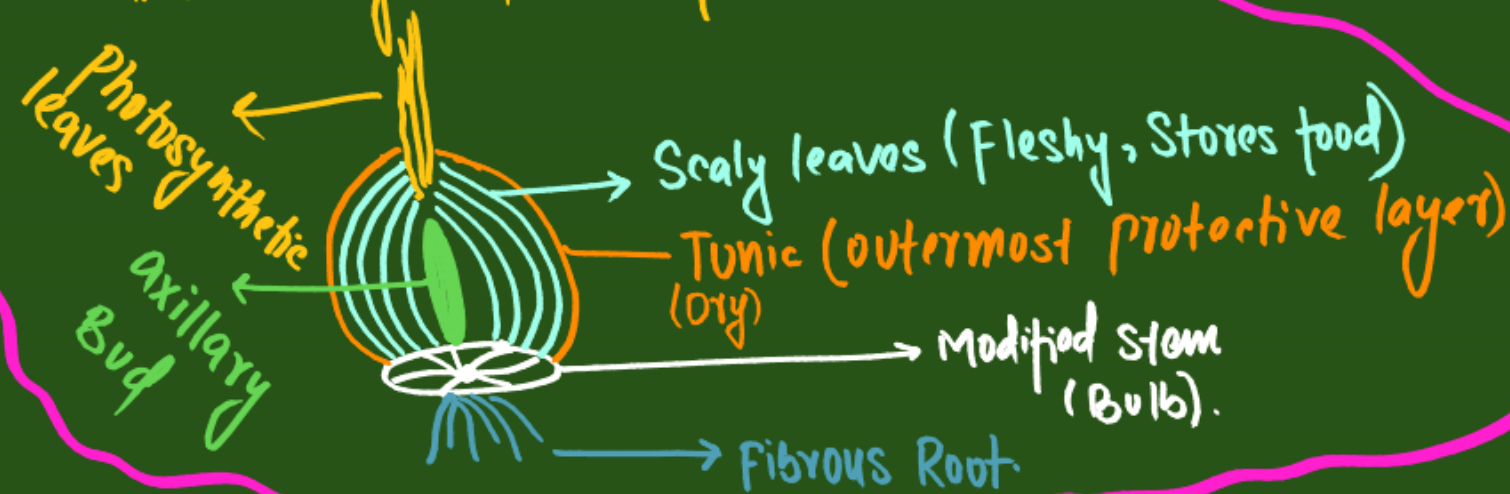
① Rhizome: Grows horizontally under Soil.
Ex. Ginger, Turmeric, Sugarcane, Banana.



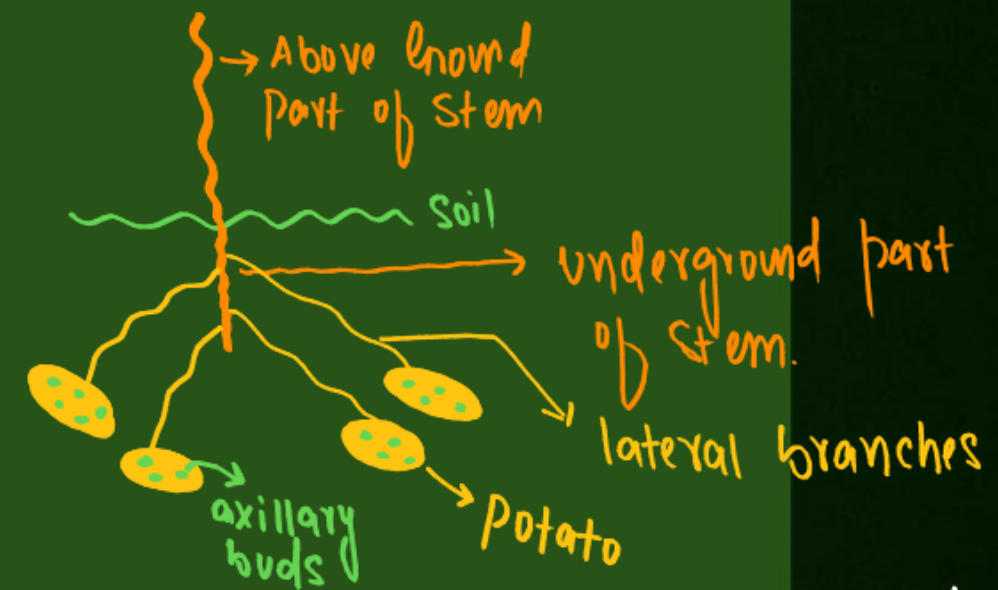
② Corm: Grows Vertically under Soil.
→ longest underground modified Stem.
Ex. Colcassia, Zaminkand, Crocus (Saffron)



④ Bulb: Ex. Onion, Garlic.
→ Smalles underground modified Stem.



③ Tuber: Ex. Potato (Solanum tuberosum)

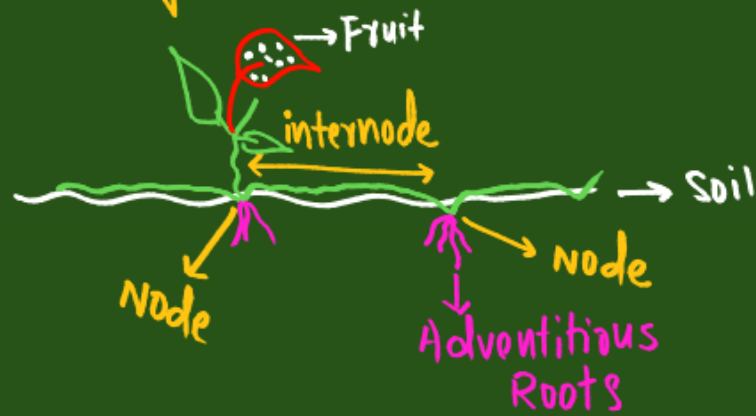


• At tips of lateral branches of underground stem - food gets stored and swells up to form potato (Tuber)
• potato bears axillary buds or eyes which helps in veg. Propagation.

(b) SUB-AERIAL STEM MODIFICATIONS

• 04 types :

① RUNNER : ex. Oxalis, Grass, Strawberry (*Fragaria*)
→ long internodes int.



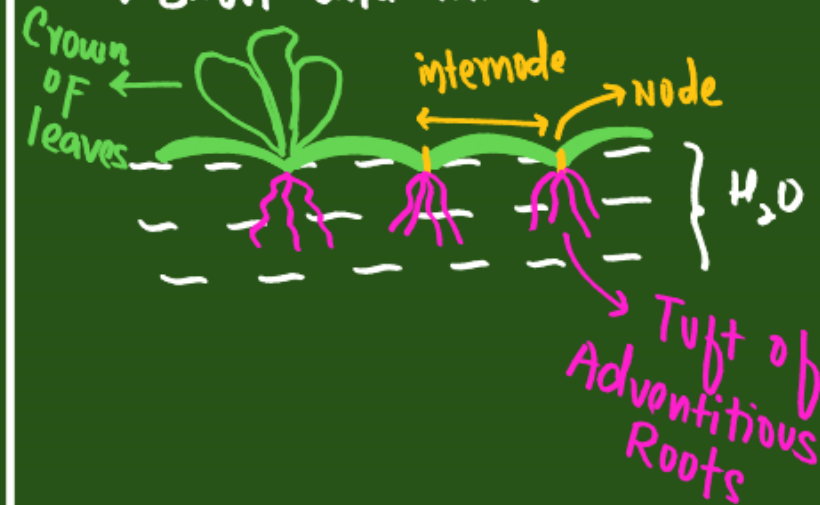
② OFFSET :

- int in Aquatic plants.
- ex. • Hyacinth (*Eicchornia*)

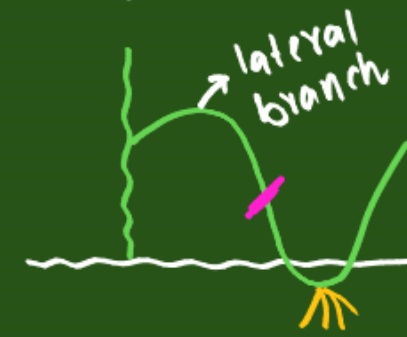
Terror of Bengal

• Pistia.

→ Short and thick internodes

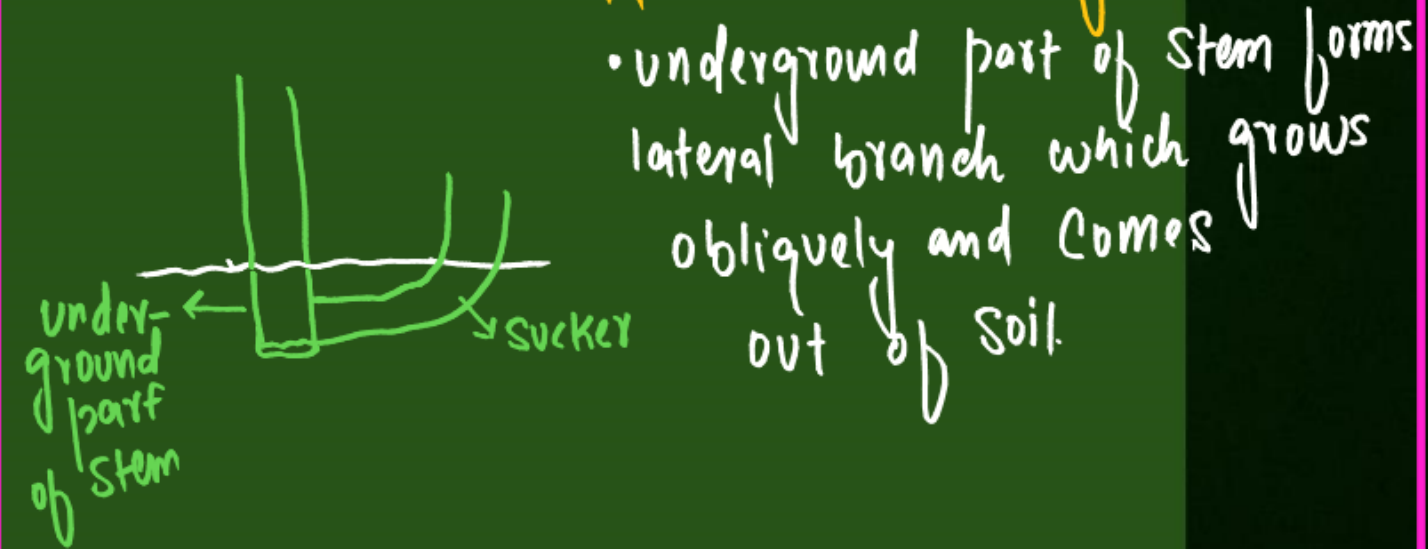


③ Stolon : ex. Jasmine, Mint



- lateral branch bends downwards and moves into the soil and then again comes out.
- lateral branch forms arches at regular intervals.

④ Sucker : ex. Pineapple, Banana, Chrysanthemum



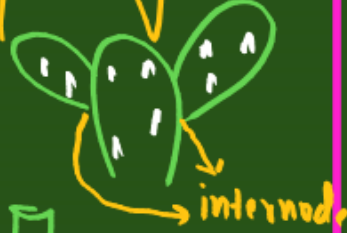
- underground part of stem forms lateral branch which grows obliquely and comes out of soil.

(C) AERIAL STEM MODIFICATIONS

- 04 types :

① Phylloclade :

- In xerophytic plants, stem becomes flattened and green — performs photosynthesis.
- Ex. Opuntia (cactus) — Flattened stem



Euphorbia — Cylindrical stem

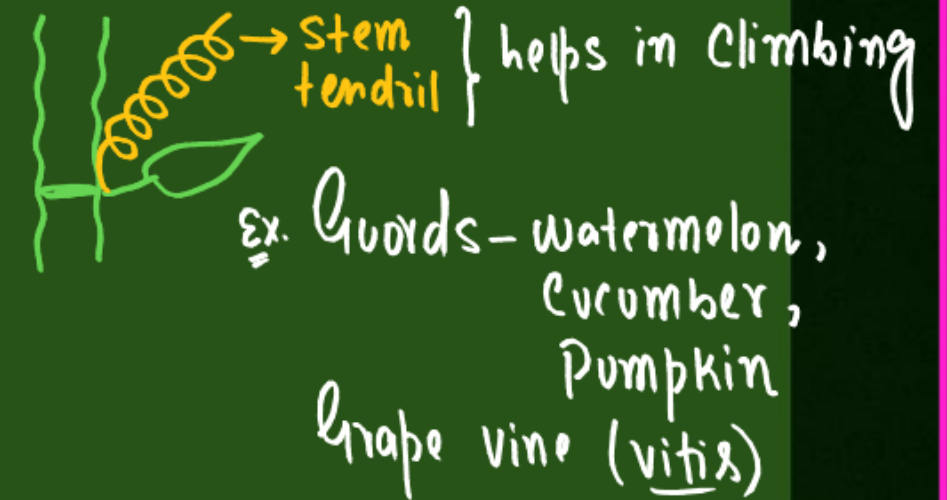


② Cladode :

- Single internode long phylloclade.
- Ex. Ruscus
Asparagus

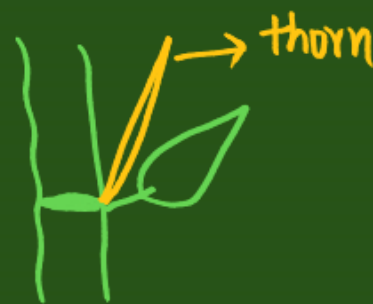
③ Stem tendril :

- Axillary buds form slender and coiled st. called as tendril



④ Thorn :

- Axillary bud modifies and forms thorn (lib)
- provides protection from browsing animals.



Ex. Citrus, Bougainvillea

QUESTIONS

In Bougainvillea, thorns are the modifications of:

(2017-Delhi)

- ☐ A Stipules
- ☐ B Adventitious root
- ☒ C Stem
- ☐ D Leaf

M/W

QUESTIONS

Which of the following statements is not correct?

(2022 Re)

- A** The rhizome is thick, prostrate and branched
- B** Rhizome is a condensed form of stem
- C** The apical bud in rhizome always remains above the ground
- D** The rhizome is aerial with no distinct nodes and internodes

QUESTIONS

u/w

Stems modified into flat green organs performing the functions of leaves are known as: (2016 - 1)

- A** Cladodes
- B** Phyllodes
- C** Phylloclades
- D** Scales

QUESTIONS

H/W



Which of the following is not a stem modification?

(2016 - I)

- A** Pitcher of Nepenthes
- B** Tendrils of cucumber
- C** Thorns of Citrus
- D** Flattened structures of Opuntia

QUESTIONS

4/10



An example of edible underground stem is:

(2014)

- A** Potato
- B** Carrot
- C** Groundnut
- D** Sweet potato

HOMEWORK

04 Questions



Thank
You